Aurora: Module Relay x8

Stakeholders Statements

Prepared for

Technical Staff

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Disclosure is prohibited

Revision History

Version	Date	Author	Description
0.1.0	2026-07-02	Martin Pellizzer	Starting version

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Statements

Statement 1

Statement

The Relay Module shall execute relay-output commands received from CORE

Answer

The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.

Question

What exactly is the responsibility of the Relay Module within Aurora?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	1
Answer ID	1
Stakeholder Role	System Architect
Question Text	What exactly is the responsibility of the Relay Module within Aurora?
Answer Text	The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.
Statement Text	The Relay Module shall execute relay-output commands received from CORE

Statement 2

Statement

The Relay Module shall provide CORE with output status

Answer

The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.

Question

What exactly is the responsibility of the Relay Module within Aurora?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	1
Answer ID	1
Stakeholder Role	System Architect
Question Text	What exactly is the responsibility of the Relay Module within Aurora?
Answer Text	The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.
Statement Text	The Relay Module shall provide CORE with output status

Statement 3

Statement

The Relay Module shall provide CORE with module diagnostics

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall provide CORE with module diagnostics

Statement 4

Statement

The Relay Module shall transition its outputs to safe states when system-level fault

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall transition its outputs to safe states when system-level fault

Statement 5

Statement

The Relay Module shall provide eight indipendently controlled relay-driven channels

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall provide eight indipendently controlled relay-driven channels

Statement 6

Statement

The Relay Module shall manage activation of each channel

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall manage activation of each channel

Statement 7

Statement

The Relay Module shall manage deactivation of each channel

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall manage deactivation of each channel

Statement 8

Statement

The Relay Module shall monitor output faults

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall monitor output faults

Statement 9

Statement

The Relay Module shall communicate channels status to CORE

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall communicate channels status to CORE

Statement 10

Statement

The Relay Module shall communicate channels diagnostics to CORE

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall communicate channels diagnostics to CORE

Statement 11

Statement

The Relay Module shall communicate to CORE via Modbus RTU protocol

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall communicate to CORE via Modbus RTU protocol

Statement 12

Statement

The Relay Module shall communicate to CORE via RS-485 hardware standard

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall communicate to CORE via RS-485 hardware standard

Statement 13

Statement

The Relay Module shall enforce local fault-handling

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Statement Text	The Relay Module shall enforce local fault-handling

Statement 14

Statement

The Relay Module shall turn ON/OFF outputs within specified time

Answer

Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.

Question

What behavior does the system require from each of the eight relay outputs?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	3
Answer ID	3
Stakeholder Role	System Architect
Question Text	What behavior does the system require from each of the eight relay outputs?
Answer Text	Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.
Statement Text	The Relay Module shall turn ON/OFF outputs within specified time

Statement 15

Statement

The Relay Module shall prevent one faulty channel to affect other channels

Answer

Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.

Question

What behavior does the system require from each of the eight relay outputs?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	3
Answer ID	3
Stakeholder Role	System Architect
Question Text	What behavior does the system require from each of the eight relay outputs?
Answer Text	Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.
Statement Text	The Relay Module shall prevent one faulty channel to affect other channels

Statement 16

Statement

The Relay Module shall maintain the output defined safe behavior under fault conditions

Answer

Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.

Question

What behavior does the system require from each of the eight relay outputs?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	3
Answer ID	3
Stakeholder Role	System Architect
Question Text	What behavior does the system require from each of the eight relay outputs?
Answer Text	Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.
Statement Text	The Relay Module shall maintain the output defined safe behavior under fault conditions

Statement 17

Statement

The Relay Module shall detect communication timeout with CORE

Answer

Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.

Question

What must happen to every relay output when communication with the CORE is lost?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	4
Answer ID	4
Stakeholder Role	System Architect
Question Text	What must happen to every relay output when communication with the CORE is lost?
Answer Text	Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.
Statement Text	The Relay Module shall detect communication timeout with CORE

Statement 18

Statement

The Relay Module shall transition to its outputs to predefined safe state when timeout with CORE

Answer

Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.

Question

What must happen to every relay output when communication with the CORE is lost?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	4
Answer ID	4
Stakeholder Role	System Architect
Question Text	What must happen to every relay output when communication with the CORE is lost?
Answer Text	Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.
Statement Text	The Relay Module shall transition to its outputs to predefined safe state when timeout with CORE

Statement 19

Statement

The Relay Module shall transition to defined safe state when in failure state

Answer

Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.

Question

What is the required system behavior if the Relay Module itself fails?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	5
Answer ID	5
Stakeholder Role	System Architect
Question Text	What is the required system behavior if the Relay Module itself fails?
Answer Text	Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.
Statement Text	The Relay Module shall transition to defined safe state when in failure state

Statement 20

Statement

The Relay Module shall prevent unintended activation when in failure state

Answer

Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.

Question

What is the required system behavior if the Relay Module itself fails?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	5
Answer ID	5
Stakeholder Role	System Architect
Question Text	What is the required system behavior if the Relay Module itself fails?
Answer Text	Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.
Statement Text	The Relay Module shall prevent unintended activation when in failure state

Statement 21

Statement

The Relay Module shall provide sufficient fault information to CORE when in failure state

Answer

Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.

Question

What is the required system behavior if the Relay Module itself fails?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	5
Answer ID	5
Stakeholder Role	System Architect
Question Text	What is the required system behavior if the Relay Module itself fails?
Answer Text	Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.
Statement Text	The Relay Module shall provide sufficient fault information to CORE when in failure state

Statement 22

Statement

The Relay Module shall execute commands deterministically

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Statement Text	The Relay Module shall execute commands deterministically

Statement 23

Statement

The Relay Module shall manage its eight physical driver outputs

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Statement Text	The Relay Module shall manage its eight physical driver outputs

Statement 24

Statement

The Relay Module shall perform local hardware diagnostics

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Statement Text	The Relay Module shall perform local hardware diagnostics

Statement 25

Statement

The Relay Module shall report its status to CORE

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Statement Text	The Relay Module shall report its status to CORE

Statement 26

Statement

The Relay Module shall report its faults to CORE

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Statement Text	The Relay Module shall report its faults to CORE

Statement 27

Statement

The Relay Module shall report the state of all eight outputs

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report the state of all eight outputs

Statement 28

Statement

The Relay Module shall report the module health

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report the module health

Statement 29

Statement

The Relay Module shall report individual channel faults

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report individual channel faults

Statement 30

Statement

The Relay Module shall report communication status

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report communication status

Statement 31

Statement

The Relay Module shall report electrical identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report electrical identification

Statement 32

Statement

The Relay Module shall report thermal identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report thermal identification

Statement 33

Statement

The Relay Module shall report diagnostics identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report diagnostics identification

Statement 34

Statement

The Relay Module shall report firmware identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report firmware identification

Statement 35

Statement

The Relay Module shall report hardware identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report hardware identification

Statement 36

Statement

The Relay Module shall report any condition preventing reliable output control

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Statement Text	The Relay Module shall report any condition preventing reliable output control

Statement 37

Statement

The Relay Module shall reliably energize its assigned relay coil across the full specified supply

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall reliably energize its assigned relay coil across the full specified supply

Statement 38

Statement

The Relay Module shall independently energize its assigned relay coil across the full specified supply

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall independently energize its assigned relay coil across the full specified supply

Statement 39

Statement

The Relay Module shall energize its assigned relay coil across the full specified temperature

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil across the full specified temperature

Statement 40

Statement

The Relay Module shall energize its assigned relay coil across the full specified tolerance range

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil across the full specified tolerance range

Statement 41

Statement

The Relay Module shall energize its assigned relay coil with adequate voltage

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with adequate voltage

Statement 42

Statement

The Relay Module shall energize its assigned relay coil with adequate voltage margin

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with adequate voltage margin

Statement 43

Statement

The Relay Module shall energize its assigned relay coil with adequate current margin

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with adequate current margin

Statement 44

Statement

The Relay Module shall energize its assigned relay coil with continuous-duty

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with continuous-duty

Statement 45

Statement

The Relay Module shall energize its assigned relay coil with continuous-duty capability

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with continuous-duty capability

Statement 46

Statement

The Relay Module shall energize its assigned relay coil with short-circuit protection

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with short-circuit protection

Statement 47

Statement

The Relay Module shall energize its assigned relay coil with overload protection

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with overload protection

Statement 48

Statement

The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions

Statement 49

Statement

The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Statement Text	The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions

Statement 50

Statement

The Relay Module shall operate with coils that have a defined nominal voltage

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have a defined nominal voltage

Statement 51

Statement

The Relay Module shall operate with coils that have allowable voltage range

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have allowable voltage range

Statement 52

Statement

The Relay Module shall operate with coils that have nominal current

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have nominal current

Statement 53

Statement

The Relay Module shall operate with coils that have maximum current

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have maximum current

Statement 54

Statement

The Relay Module shall operate with coils that have defined resistance

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have defined resistance

Statement 55

Statement

The Relay Module shall operate with coils that have defined inductance

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have defined inductance

Statement 56

Statement

The Relay Module shall operate with coils that have defined inrush behavior

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have defined inrush behavior

Statement 57

Statement

The Relay Module shall operate with coils that have defined continuous-duty rating

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have defined continuous-duty rating

Statement 58

Statement

The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

Statement 59

Statement

The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Statement Text	The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

Statement 60

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range

Statement 61

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances

Statement 62

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation

Statement 63

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions

Statement 64

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts

Statement 65

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage

Statement 66

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Statement 67

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

Statement 68

Statement

The Relay Module shall have drivers with appropriately sized suppressions

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Statement Text	The Relay Module shall have drivers with appropriately sized suppressions

Statement 69

Statement

The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Statement Text	The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode

Statement 70

Statement

The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Statement Text	The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

Statement 71

Statement

The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Statement Text	The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings

Statement 72

Statement

The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Statement Text	The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays

Statement 73

Statement

The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

Answer

During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.

Question

What electrical behavior is required during power loss, brownout, and restoration of power?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	15
Answer ID	15
Stakeholder Role	Electrical Engineer
Question Text	What electrical behavior is required during power loss, brownout, and restoration of power?
Answer Text	During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.
Statement Text	The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

Statement 74

Statement

The Relay Module shall transition all relay outputs to defined safe state during power loss

Answer

During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.

Question

What electrical behavior is required during power loss, brownout, and restoration of power?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	15
Answer ID	15
Stakeholder Role	Electrical Engineer
Question Text	What electrical behavior is required during power loss, brownout, and restoration of power?
Answer Text	During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.
Statement Text	The Relay Module shall transition all relay outputs to defined safe state during power loss

Statement 75

Statement

The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

Answer

During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.

Question

What electrical behavior is required during power loss, brownout, and restoration of power?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	15
Answer ID	15
Stakeholder Role	Electrical Engineer
Question Text	What electrical behavior is required during power loss, brownout, and restoration of power?
Answer Text	During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.
Statement Text	The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

Statement 76

Statement

The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Statement Text	The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels

Statement 77

Statement

The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Statement Text	The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels

Statement 78

Statement

The Relay Module shall detect relevant hardware faults

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Statement Text	The Relay Module shall detect relevant hardware faults

Statement 79

Statement

The Relay Module shall detect relevant output faults

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Statement Text	The Relay Module shall detect relevant output faults

Statement 80

Statement

The Relay Module shall transition affected outputs to their predefined safe states independently of higher-level system decisions

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Statement Text	The Relay Module shall transition affected outputs to their predefined safe states independently of higher-level system decisions

Statement 81

Statement

The Relay Module shall have a formally defined safe state for all its outputs based on its associated hazard analysis

Answer

Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.

Question

What is the required safe state for each of the eight relay outputs?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	17
Answer ID	17
Stakeholder Role	Safety Engineer
Question Text	What is the required safe state for each of the eight relay outputs?
Answer Text	Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.
Statement Text	The Relay Module shall have a formally defined safe state for all its outputs based on its associated hazard analysis

Statement 82

Statement

The Relay Module shall have OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized

Answer

Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.

Question

What is the required safe state for each of the eight relay outputs?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	17
Answer ID	17
Stakeholder Role	Safety Engineer
Question Text	What is the required safe state for each of the eight relay outputs?
Answer Text	Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.
Statement Text	The Relay Module shall have OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized

Statement 83

Statement

The Relay Module shall immediately prevent unsafe output activation when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Statement Text	The Relay Module shall immediately prevent unsafe output activation when it detects an internal fault capable of compromising safe operation

Statement 84

Statement

The Relay Module shall force all affected outputs to their defined safe states when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Statement Text	The Relay Module shall force all affected outputs to their defined safe states when it detects an internal fault capable of compromising safe operation

Statement 85

Statement

The Relay Module shall latch the fault as required when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Statement Text	The Relay Module shall latch the fault as required when it detects an internal fault capable of compromising safe operation

Statement 86

Statement

The Relay Module shall report the fault as required when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Statement Text	The Relay Module shall report the fault as required when it detects an internal fault capable of compromising safe operation

Statement 87

Statement

The Relay Module shall prevent reactivation until the fault-clear conditions are satisfied when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Statement Text	The Relay Module shall prevent reactivation until the fault-clear conditions are satisfied when it detects an internal fault capable of compromising safe operation

Statement 88

Statement

The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Statement Text	The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

Statement 89

Statement

The Relay Module shall autonomously transition every output to its predefined safe state when communication with the CORE is lost beyond the defined safety timeout

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Question

What shall happen to each relay output when communication with the CORE is lost?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.
Statement Text	The Relay Module shall autonomously transition every output to its predefined safe state when communication with the CORE is lost beyond the defined safety timeout

Statement 90

Statement

The Relay Module shall maintain a predefined safe state when communication with the CORE is lost beyond the defined safety timeout

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Question

What shall happen to each relay output when communication with the CORE is lost?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.
Statement Text	The Relay Module shall maintain a predefined safe state when communication with the CORE is lost beyond the defined safety timeout

Statement 91

Statement

The Relay Module shall report the communication fault when communication with CORE is restored

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Question

What shall happen to each relay output when communication with the CORE is lost?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.
Statement Text	The Relay Module shall report the communication fault when communication with CORE is restored

Statement 92

Statement

The Relay Module shall require explicit reauthorization from CORE before returning to normal operation

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Question

What shall happen to each relay output when communication with the CORE is lost?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.
Statement Text	The Relay Module shall require explicit reauthorization from CORE before returning to normal operation

Statement 93

Statement

The Relay Module shall locally enforce output-level safety constraints

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.
Statement Text	The Relay Module shall locally enforce output-level safety constraints

Statement 94

Statement

The Relay Module shall locally enforce watchdog/communication-loss responses

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.
Statement Text	The Relay Module shall locally enforce watchdog/communication-loss responses

Statement 95

Statement

The Relay Module shall locally enforce internal fault handling

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.
Statement Text	The Relay Module shall locally enforce internal fault handling

Statement 96

Statement

The Relay Module shall locally enforce defined safe-state behavior without relying solely on the CORE

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.
Statement Text	The Relay Module shall locally enforce defined safe-state behavior without relying solely on the CORE

Statement 97

Statement

The Relay Module shall manage the eight relay outputs to control or influence the ozone-generation process by operating the required process equipment and auxiliaries

Answer

The eight relay outputs shall control or influence the ozone-generation process by operating the required process equipment and auxiliaries—such as ozone-generation power, cooling, gas/air or oxygen supply, valves, purge/ventilation, and other process actuators—according to the defined process sequence and interlocks.

Question

What process functions are controlled or influenced by the eight relay outputs of the Relay Module?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	21
Answer ID	21
Stakeholder Role	Process Engineer
Question Text	What process functions are controlled or influenced by the eight relay outputs of the Relay Module?
Answer Text	The eight relay outputs shall control or influence the ozone-generation process by operating the required process equipment and auxiliaries—such as ozone-generation power, cooling, gas/air or oxygen supply, valves, purge/ventilation, and other process actuators—according to the defined process sequence and interlocks.
Statement Text	The Relay Module shall manage the eight relay outputs to control or influence the ozone-generation process by operating the required process equipment and auxiliaries

Statement 98

Statement

The Relay Module shall include a startup sequence that verifies all required permissives

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Statement Text	The Relay Module shall include a startup sequence that verifies all required permissives

Statement 99

Statement

The Relay Module shall include a startup sequence that establishes the required gas flow

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Statement Text	The Relay Module shall include a startup sequence that establishes the required gas flow

Statement 100

Statement

The Relay Module shall include a startup sequence that establishes the supporting utilities

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Statement Text	The Relay Module shall include a startup sequence that establishes the supporting utilities

Statement 101

Statement

The Relay Module shall include a startup sequence that enables cooling

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Statement Text	The Relay Module shall include a startup sequence that enables cooling

Statement 102

Statement

The Relay Module shall include a startup sequence that enables other auxiliaries

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Statement Text	The Relay Module shall include a startup sequence that enables other auxiliaries

Statement 103

Statement

The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Statement Text	The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries

Statement 104

Statement

The Relay Module shall include a startup sequence that energizes the ozone-generation system

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Statement Text	The Relay Module shall include a startup sequence that energizes the ozone-generation system

Statement 105

Statement

The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Statement Text	The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete

Statement 106

Statement

The Relay Module shall include a shutdown sequence that disables ozone generation

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Statement Text	The Relay Module shall include a shutdown sequence that disables ozone generation

Statement 107

Statement

The Relay Module shall include a shutdown sequence that maintains the required cooling

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Statement Text	The Relay Module shall include a shutdown sequence that maintains the required cooling

Statement 108

Statement

The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Statement Text	The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period

Statement 109

Statement

The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Statement Text	The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order

Statement 110

Statement

The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Statement Text	The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

Statement 111

Statement

The Relay Module shall permit ozone generation only when all required utilities are available

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when all required utilities are available

Statement 112

Statement

The Relay Module shall permit ozone generation only when gas flow is within their specified limits

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when gas flow is within their specified limits

Statement 113

Statement

The Relay Module shall permit ozone generation only when cooling is within their specified limits

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when cooling is within their specified limits

Statement 114

Statement

The Relay Module shall permit ozone generation only when required valves are in their correct states

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when required valves are in their correct states

Statement 115

Statement

The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states

Statement 116

Statement

The Relay Module shall permit ozone generation only when no active safety interlock is violated

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when no active safety interlock is violated

Statement 117

Statement

The Relay Module shall permit ozone generation only when no active process interlock is violated

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when no active process interlock is violated

Statement 118

Statement

The Relay Module shall permit ozone generation only when all required feedback signals are valid

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when all required feedback signals are valid

Statement 119

Statement

The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Statement Text	The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

Statement 120

Statement

The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Statement Text	The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected

Statement 121

Statement

The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Statement Text	The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow

Statement 122

Statement

The Relay Module shall immediately stop ozone generation when detected a loss of required cooling

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Statement Text	The Relay Module shall immediately stop ozone generation when detected a loss of required cooling

Statement 123

Statement

The Relay Module shall immediately stop ozone generation when detected a critical equipment failure

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Statement Text	The Relay Module shall immediately stop ozone generation when detected a critical equipment failure

Statement 124

Statement

The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Statement Text	The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits

Statement 125

Statement

The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Statement Text	The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives

Statement 126

Statement

The Relay Module shall immediately stop ozone generation when detected emergency shutdown

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Statement Text	The Relay Module shall immediately stop ozone generation when detected emergency shutdown

Statement 127

Statement

The Relay Module shall immediately stop ozone generation when detected defined critical interlock

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Statement Text	The Relay Module shall immediately stop ozone generation when detected defined critical interlock

Statement 128

Statement

The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Statement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple